

# Traffic Records Improvement Program Reporting System (TRIPRS)

## Statement of Work

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*Version 1.0*

Prepared by

The U.S. Department of Transportation  
National Highway Traffic Safety Administration  
Office of National Driver Register and Traffic Records Division  
(NDR&TR)

# **1 INTRODUCTION**

The National Highway Traffic Safety Administration (NHTSA) is an agency of the U.S. Department of Transportation (DOT). NHTSA's mission is to save lives, prevent injuries and reduce traffic-related health care and other economic costs. The agency develops, promotes and implements effective educational, engineering and enforcement programs with the goal of ending preventable tragedies and reducing economic costs associated with vehicle use and highway travel. NHTSA accomplishes this in part by awarding State Traffic Safety Information System Improvement grants to support State efforts to improve their data systems. NHTSA's National Driver Register and Traffic Records Division (NDR&TR) is tasked with helping states improve their traffic safety data collection, management, and analysis capabilities through evaluation, training, and technical assistance. The NDR&TR Division is also responsible for the Model Minimum Uniform Crash Criteria (MMUCC) and helping states align their crash systems and documentation to the latest edition of MMUCC.

## **2 BACKGROUND**

NHTSA's NDR&TR support for States centers on six traffic records data systems (Crash, Driver, Vehicle, Roadway, Citation/Adjudication, Injury Surveillance). NHTSA commissioned the development of the Traffic Records Improvement Program Reporting System (TRIPRS) in 2010 to monitor and track the advances in the States' Traffic Safety Information Systems technology and practices. TRIPRS allows the collection and organization of data and currently houses State documents and information collected in the normal operations of the State Traffic Records Assessment Program (STRAP); the Safety Program Assessment (SPA) module that currently includes assessments for pedestrian and bicycle safety, impaired driving, and occupant protection programs; the Data Mapping module; and the State Electronic Data Collection grant application and monitoring module.

The NHTSA Office of the Chief Information Officer (OCIO) operates and maintains a set of virtualized Windows servers that host various Traffic Records data systems. Currently there are three pairs of servers that are supporting the TRIPRS application. Each pair is a combination of a database/application server and a web server. One pair is for disaster recovery, one pair is for development and testing, and one pair is the production system host. Each pair utilizes the latest version of Oracle Enterprise database 19.7.0.0.0 (19c), Oracle Application Express (21.2.0.0.0), Oracle ORDS (21.4.0.0.0), Oracle Business Intelligence (BI) Publisher (12.2.1.4.0), Oracle Web Logic (12.2.1.4.0) (Web Application Server), and Oracle Web Logic (12.2.1.4.0) (BI Publisher Hosting Server).

## **3 OBJECTIVES**

The purpose of this project is to support develop, enhance Traffic Records Improvement Program Reporting System (TRIPRS) system while sustaining the software in the Production environment. The primary goal is to O&M Support, enhance and expand existing features, provide backend integration with Oracle cloud Infrastructure (OCI) services and other NHTSA systems. Note that this includes non-functional requirements, which include optimizing the development environment, cost control, optimization, performance and user experience, all of which must be addressed in lockstep with functional requirements. Continuous improvement is a necessity and a key common thread across DOT projects. Thus, the architecture and design principles utilized for

TRIPRS system, as well as the software development process and associated tooling, must also continue to be refined, enhanced and remain extensible to facilitate seamless implementation of future requirements.

Across all programs and projects, using Agile methods and sound software engineering best practices, DOT generally seeks to leverage automation to reduce manual efforts, increase efficiencies, reduce operational risks, and reduce overall costs by implementing self-service features, automating manual tasks and increasingly incorporating new and emerging technologies, such as Artificial Intelligence (AI) and Machine Learning (ML) in addition to existing push for serverless and managed cloud services whenever possible. Additionally, as all S93 digital factory programs and projects contribute to the same DOT mission, they often operate on closely related data. Thus, integration and cross-referencing of information between IT solutions within DOT is important.

The automation goal in particular also includes the software development lifecycle itself. Like many projects at DOT, It is the goal of DOT to ensure all stages of the CI/CD pipeline are sufficiently automated such that continuous delivery, comprised of any number of small changes, are deployed to the Production environment frequently without manual intervention.

Additionally, in alignment with a key guidance from the S93 Digital factory Chief Technology Officer (CTO), all implementations, regardless of which project they original from, must take not only a cloud-first but also an enterprise-first approach so that the solutions can be deployed across DOT and be readily re-usable by other projects within DOT.

Using these guiding principles – automation, self-service, enterprise-first, leveraging emerging technologies and continuous improvement, the goal of the TRIPRS project to provide users with a modern, high-quality software solution, built with highly automated modern techniques, that offers exceptional user experience which allows them to carry out their mission activities with ease, efficiency and reduced cost. Along the same guiding principles, the Contractor is encouraged to be innovative and make recommendations for improvements in processes, techniques, tools and new technologies.

The work performed through this contract will be coordinated with the Traffic Records (TR) Project Manager, Contracting Officer Representative (COR) (if different), the DOT OCIO, and other contractor staff. Work under this contract will provide TRIPRS application and Oracle software support. TRIPRS user support (help desk and training) will be contracted separately from this solicitation.

## **4 SCOPE OF WORK**

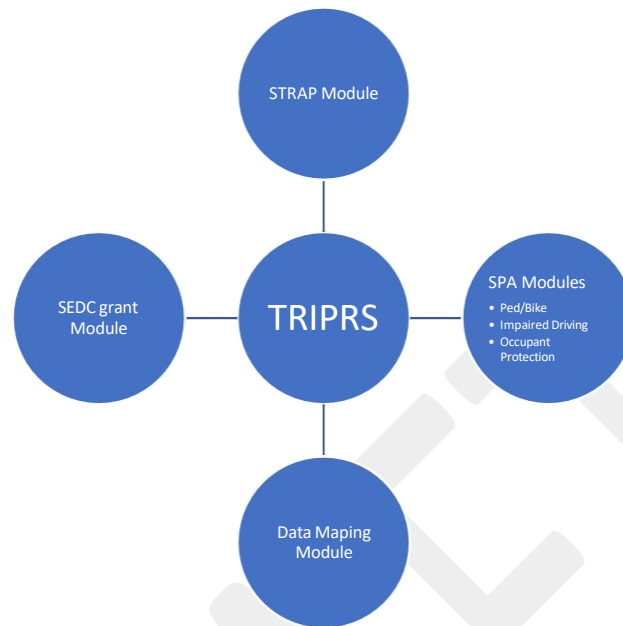
The scope of this statement of work relates Operations and Maintenance (O&M) and Optional Development, Modernization and Enhancement (DME) support for the Traffic Records Improvement Program Reporting System (TRIPRS) for the DOT's National Driver Register and Traffic Records Division (NDR&TR). TRIPRS is hosted on an Oracle Cloud Infrastructure (OCI) Platform.

### **4.1 SPECIFIC REQUIREMENTS**

#### **4.1.1 Operations And Maintenance of the TRIPRS Application**

The contractor shall provide Operations and Maintenance (O&M) support (e.g. Application support, Database management, documentation, reports, authorization to operate (ATO)) to the current TRIPRS application. The TRIPRS application currently includes STRAP, SPA, Data

Mapping, SEDC grant monitoring module, a Document Library (used by all modules), and a Sandbox module. All modules are accessible through TRIPRS and several have linked Tableau data dashboards.



*Figure 1: TRIPRS and Existing, Connected Modules (STRAP, SPA, Data Mapping, SEDC, Document library and Sandbox)*

#### **4.1.2 Development, Modernization, and Enhancement (DME) → These are Optional tasks based on funds availability**

As an actively used system to support the mission of NHTSA's Traffic Records Division activities, the TRIPRS system will have an evolving backlog containing various requirements and system enhancements. In accordance with the Agile approach, should the Government award this optional task, the Contractor shall lead efforts under the direction of the COR to groom and refine these Product Backlog items on a biweekly basis, or as directed by the COR. As part of the refinement activities, the Contractor shall capture new requirements, conduct discovery and elicitation, and track them using NHTSA's Agile project management tools, prioritizing them as directed by the COR. As described in the Software Development Process section of this document, each item in the Product Backlog must be sufficiently detailed and complete, and the Contractor shall generally maintain a clean, accurate, and up-to-date Product Backlog. The Contractor shall lead collaboration and coordination efforts with other DOT/NHTSA teams, as appropriate, analyze relevant requirements, develop and document detailed requirements, and design, develop, and deploy capabilities that meet these requirements. The Contractor shall provide work products, milestones, and deliverables for new requirements in accordance with the Agile schedule. This means that the deliverable is generally due and updated on cadence, every Sprint or Release, or as directed by the COR. The Contractor may negotiate the specifics of each milestone and deliverable with the COR.

#### 4.1.2.1 *Develop new State highway safety program assessment sub-module within SPA (Optional)*

The contractor shall leverage the existing program assessment sub-module format in the SPA module (i.e., Impaired Driving, Occupant Protection, Ped/Bike) to build new State highway safety program assessment sub-modules (e.g., motorcycle program, EMS program). This includes replicating the highway safety program assessment questions within the sub-module, testing, validating, deploying, and documenting using Agile process.

#### 4.1.2.2 *TRIPRS Modernizing (Optional)*

The TRIPRS system has an evolving backlog item containing various bugs and system enhancements to already implemented capabilities detailed below under Fixes and Enhancements, Text and Tasks. The contractor shall include these backlog items part of TRIPRS Modernization task. The Modernizing plan should include Proof of concept and other details, from the current client server architecture to a DOT approved Oracle cloud environment.

##### ➤ **Fixes and Enhancements:**

- Identify Respondents to match SPA functionality. Add pick list and ability to Limit to assessment state
- There is an "easy button" for the SPA Team Leader/Editor role. Can we add this ability to Assessors in STRAP? Can we also add this for the SPA modules?
- TRIPRS currently logs the dates and recipient of system-generated emails, but not the content. They could start tracking the content as well, but this would be a heavy lift. It could be helpful for verifying that links are valid and working.
- Updates to TRIPRS to automate report templates for customized assessments is a heavy lift—a good time to explore this would be when TRIPRS moves to a new platform.
- Please create a report that can count how many times a question has been used in recommendations
- Eliminate codes for respondents in the assessment report--just use respondent names instead of their codes

##### ➤ **Text**

- System Access
- Token
- Help tab on Left –All sections
- Verify the Language and links
- Update the "What do I do" sections, roles have changed and contact info. Also note that the text changes font.
- Change page reference text from "Page" to "Page(s)"

##### ➤ **Tasks**

- Develop a Modernization plan based on current system functionalities align with Oracle Cloud plat form and DOT Standards including Google Work space **Capabilities.**
- Review Modernized plan with Program office team and DOT Enterprise Architecture team to align DOT Standards.

- After receiving an approval from Program office and DOT EA teams. Contractor shall work with Program office team for planned delivery including Data migration, reports and any other third-party software products.

#### 4.1.2.3 *Model Minimum Uniform Crash Criteria (MMUCC) Mapping AI Project Integration (Optional)*

The NHTSA teams are developing a stand-alone MMUCC mapping AI tool using Databricks. The contractor shall submit a proposed Integration plan for building TRIPRS Integration with this MMUCC mapping AI tool. The plan should include Proof of concept, planned delivery schedules, and AI capabilities using Agile process.

##### ➤ **Fixes and Enhancements:**

- Input a spreadsheet to create State data structure and mapping report
- When creating a State structure, we don't want the page to have to reload every time we click on something new.
- Alphabetize the elements in the State structure for easier reading
- scrolling through the "MMUCC Active Mappings" section could be improved.

##### ➤ **Integration**

- Integrate EDMAS MMUCC mapping w/ TRIPRS

## 4.2 **Technical Requirements**

The TRIPRS system is hosted in Oracle Cloud Infrastructure (OCI) and the environment is Federal Risk and Authorization Management Program (FedRAMP) certified at the Moderate level. Contractor shall continue to make use of these and similar managed services and approaches, especially as they improve over time, for all development activities under this SOW. The Contractor shall ensure that deprecated features in OCI or any oracle component of the technology stack are resolved well in advance of obsolesce.

The Contractor shall ensure the system makes use of reusable components and provides a consistent UI behavior across all aspects of the application. The Contractor shall ensure that all UI designs adhere to NHTSA website style guidelines, use styles, layout, color scheme, and fonts approved by NHTSA. The Contractor shall ensure that the web application adheres to the NHTSA Human Interaction Guidelines (HIG), and the Contractor shall design and develop easy-to-use web application interfaces for all new features, enhancements, and applications. The Contractor shall ensure that all services and applications are Section 508 compliant per DOT policy.

The Contractor shall follow well-established conventions and apply sensible defaults across the system, especially where explicit guidance is not provided by NHTSA. Under no circumstances shall the Contractor use hard-coded values.

The Contractor shall ensure consistent behavior for similar UI controls and components across the application. The Contractor shall ensure that all UI elements and controls are highly responsive with a response time of 1 second to refresh content as a result of user actions, except where the COR has explicitly approved longer responses times for highly specialized, infrequently used functionality based on the Contractor's request. Even in such

exceptional cases, which must be few, the Contractor shall ensure the maximum response time does not exceed 8 seconds.

In general, the Contractor shall ensure that all UI elements provide options to show and/or link to data associated to the displayed information, so that users can quickly access all related data and documents. Where appropriate, the Contractor shall provide UI controls that allow the user to create associations between entities. The Contractor shall provide the users with automatic recommendations for potentially related data based on a number of criteria that NHTSA will provide.

The Contractor shall provide a highly positive user experience across these systems by implementing usable functionality with intuitive, conventional user interface components and the effective realization of the workflows and business rules. The Contractor shall incorporate extensible designs that easily facilitate future enhancements, maximize user productivity and efficiency, maximize system performance and responsiveness, and enable efficient maintenance of the system.

The Contractor is expected to apply their skills and experience in developing modern, performant and highly usable cloud hosted systems. As an innovation and continuous improvement, the Contractor is expected to stay current with the evolving technology landscape and encouraged to not only leverage DOT/NHTSA approved technologies and tools within and in the development of the TRIPRS systems, but also make recommendations for new ones for utilization within the TRIPRS and NHTSA.

#### **4.2.1 Authentication and Authorization**

The Contractor shall identify and implement the access control permissions required for each new feature or enhancement.

For internal users, the Contractor shall ensure that only Multi-factor Authentication (MFA), currently using the DOT issued Personal Identity Verification (PIV), is allowed.

For external users using the enterprise portal, the Contractor shall ensure that only MFA is allowed, and that the MFA is with a trusted identity provider, which is currently Login.gov.

The Contractor shall ensure that an authenticated user's activities are logged and the logged in session is maintained across different tabs/windows of the browser. The Contractor shall ensure the authorization mechanism extends to all objects and services touched in the course of executing a user request or action to ensure proper end-to-end authorization.

#### **4.2.2 Application Performance Monitoring and Alerting**

The Contractor shall always ensure total view of the health of the TRIPRS systems across the entire technology stack by ensuring the TRIPRS system is highly observable and setting up continuous monitoring of the system using NHTSA's observability and application performance monitoring tools and services (NHTSA approved Application Performance Monitoring (APM) tools, such as New Relic). The Contractor shall ensure that monitoring spans not only the health of individual services and events but the end-to-end application request/response cycles of the web applications, covering all layers of the application/technology stack, from the browser to content delivery services to the backend microservices, API calls and database. The Contractor shall ensure that the system monitoring provides details on how well the system is performing in real-time in order to provide visibility and key performance indicators about the users' interaction with the

system both in the UI and in areas not visible to the user. The Contractor shall ensure that logs, metrics, traces and data related to the performance and monitoring are easily accessible and published to the development and operations team as well as other NHTSA stakeholders.

Additionally, the Contractor shall set up a mechanism to automatically monitor the system from a user's perspective by frequently logging into the system and exercising key functions. For example, by setting up an automated Selenium test suite.

The Contractor shall set up a set of automated and real-time alerts which are triggered when pre-defined performance parameters go out of compliance so that the contractor's team and other COR identified personnel (typically, operations team members) can review and take remedial actions as needed.

To facilitate application monitoring, the Contractor shall ensure all UI events, API calls, database queries, search queries, and in general any application-initiated actions are logged with meaningful data and easily accessible by NHTSA stakeholders. To assist users with reviewing the history of changes to all system entities they can edit, the Contractor shall support an audit trail feature that shows changes to every field of a user editable entity, including the author of the change, date and time, the old value and the new value. Such a model for audit trails is already implemented in other existing NHTSA projects.

#### **4.2.3 Cybersecurity and Privacy**

The Contractor shall ensure that the TRIPRS systems architecture always adheres to Federal, DOT, and NHTSA cybersecurity and privacy requirements. The Contractor shall ensure that cybersecurity and privacy prerequisites are prominently included as an integral, first-class concern in all phases and aspects of the Contractor's development and support activities, including technical approach, software development and IT maintenance activities. The Contractor shall ensure access to the TRIPRS systems and data are maximally secured and protected. The Contractor shall ensure that data at rest as well as in transit within the TRIPRS systems and between the TRIPRS systems and other systems is encrypted. The Contractor shall create or update, as appropriate, cybersecurity related artifacts in accordance with NHTSA cybersecurity requirements.

#### **4.2.4 Artificial Intelligence Integration**

The Contractor shall address NHTSA's evolving data analytics needs by incorporating techniques and tools from the Artificial Intelligence (AI) arena, including Machine Learning (ML) and Natural Language Processing (NLP). As part of the TRIPRS system requirements elicitations, as the need arises, the Contractor shall explore these technologies to see if and how they can be leveraged to address certain NHTSA needs with respect to the TRIPRS data. Further, the Contractor shall also consider AI-powered solutions to apply to the software development process itself, including, for example, GitHub Copilot and other similar emerging technologies. Note that only DOT/NHTSA approved technologies can be utilized at NHTSA. Therefore, the Contract must obtain written approval from the COR when leveraging new technologies within the TRIPRS systems itself or in the construction of the TRIPRS systems software.

Additionally, the Contractor shall collaborate and coordinate with other teams within NHTSA who are building the NHTSA data analytics environment using new and emerging technologies.

#### **4.2.5 Cloud Cost Control and Optimization**

The Contractor shall ensure NHTSA's use of TRIPRS cloud resources optimizes NHTSA's cost by periodically reviewing their utilization against their cost. As such, the Contractor shall develop a Cloud Cost Control & Optimization Plan (CCC&OP) and submit it to the COR on a monthly basis. Within ten (10) days of receipt, the COR will accept or reject the CCC&OP. The Contractor shall maintain and/or revise the CCC&OP as directed by the COR.

In the CCC&OP, the Contractor shall evaluate historical TRIPRS cloud resource utilizations through the most recent past month vs. incurred and projected costs and present recommendations to the COR for maximizing the value of cloud resources and minimizing costs by reducing, eliminating or consolidating unused or minimally used resources. The Contractor shall execute the COR approved recommendations immediately and ensure they are reflected in the following month's CCC&OP. In the process, the Contractor shall collaborate and coordinate with the DOT Enterprise Cloud Service (ECS) team as appropriate.

#### **4.2.6 Application Help & User Guides**

The Contractor shall develop and maintain separate User Guides for any newly developed tasks for the internal and external TRIPRS applications. Within ten (10) days of receipt, the COR will accept or reject the User Guides. The Contractor shall maintain and/or revise the User Guides as directed by the COR.

The Contractor shall ensure the User Guides are current and updated with each software release of the TRIPRS systems as applicable. The Contractor shall ensure users are able to download a bookmarked PDF of the User Guides. The Contractor shall ensure that the User Guides are well-written, are complete and include sufficient descriptions of the business workflows, business rules and user roles/access control applicable to the associated application. Where necessary, the Contractor shall ensure that help pages have screenshots and callouts to illustrate and explain the application functionality. Also as necessary, the Contractor shall provide example-driven guides and instructions to walk the reader through a given functionality.

The Contractor shall also ensure the application online help contents include clear messages and tooltips within the application to guide the users on the user interface form fields, form submissions and error messages within the application. The Contractor shall ensure that every failed operation results in a clear and concise message to the user and provides the user with a clear explanation for how to overcome the error condition.

As needed Contractor team shall work with other teams based on program office or S93 office guidance.

All the documentation must be compatible with MS, Google, and PDF formats.

### **4.3 Software Development Process**

For all software development activities (developing a new feature, enhancing existing features or fixing defects), the Contractor shall follow industry best practices in Agile Software Development Lifecycle (ASDLC) and software engineering processes. The Contractor shall develop and submit to the COR an ASDLC Plan (ASDLCP) within thirty (30) days of award. Within ten (10) days of receipt, the COR will accept or reject the ASDLCP. The Contractor shall maintain and/or revise the ASDLCP as directed by the COR.

The ASDLCP must include the end-to-end, comprehensive approach the Contractor will utilize for software development using an Agile framework to deliver capabilities and value to NHTSA incrementally, encompassing the complete lifecycle activities that include, but may not be limited to, requirements analysis, design, security considerations, development, testing, security testing, user acceptance testing (UAT) and deployment in each iteration. The Contractor is encouraged to utilize well-established programming models and paradigms, such as Test-Driven Development (TDD), Behavior-Driven Development (BDD), Pair Programming, etc. The Contractor shall ensure that the ASDLCP also defines the timelines for all software releases, Sprints, milestones, work products and deliverables which the Government has specified to be delivered according to the Agile Schedule. All ASDLCP related artifacts are considered living and the Contractor shall ensure they are kept current and submitted to the COR for review at least once prior to each software release until the end of the period of performance.

The Contractor shall rigorously incorporate into the ASDLCP the processes and artifacts described in the following sections. The Contractor shall utilize the ASDLCP as part of its overall project management plan to consistently deliver high-quality TRIPRS capabilities and feature sets that address NHTSA's needs in an iterative manner. The Contractor shall employ rigorous requirements analysis/elicitation, create fully bidirectional Requirements Traceability Matrix (RTM) and implement full automation to build, test and deploy the software solution according to well-established modern DevSecOps processes and modern software development best practices.

The Contractor shall provide and utilize the artifacts identified in this SOW and maintain all of them throughout the period of performance. Not all artifacts may be required for all efforts. As appropriate for a given effort, the Contractor shall request approval from the COR to exclude artifacts that are not applicable to that effort. Unless the Contractor obtains written approval from the COR for exclusions, the Contractor shall develop and maintain all artifacts outlined in this SOW. The Contractor shall submit each artifact for review and approval by the COR in accordance with the Agile Schedule. The COR will review and return the artifact to the Contractor with comments and edits or approve the artifact in accordance with the Agile Schedule.

Many of the milestones and deliverables the Contractor is required to provide for this effort will indicate their schedule is in accordance with the "Agile Schedule". This means that the deliverable is generally due and updated on cadence, every Sprint or Release, or as directed by the COR. The Contractor may negotiate the specifics of each milestone and deliverable with the COR. The Contractor shall include the finalized, COR directed schedule in the ASDLCP.

#### **4.3.1 Agile Process**

The Contractor shall have experience and expertise developing functional, high-quality mission critical enterprise software and web applications using Agile frameworks and state-of-the-art in modern software engineering techniques and tools. Thus, the Contractor shall adapt to continually evolving requirements, changing compliance mandates and rapidly advancing technology landscape.

The Contractor shall develop all software within the scope of this SOW using the Scrum Agile framework using NHTSA's Agile management tools, currently Atlassian Jira and Confluence. The Contractor shall conduct Agile ceremonies or events, and develop, maintain and provide all Agile artifacts as deliverables. The table below provides a minimum set of Agile events and deliverables which the Contractor must conduct and deliver, respectively, at the specified cadence.

The COR may direct the Contractor to conduct additional events and/or provide additional deliverables or work products.

The COR will review each artifact and accept or reject it in accordance with the Agile Schedule. The Contractor shall maintain and/or revise each artifact as directed by the COR.

| <b>Event/Deliverable</b>                 | <b>Description</b>                                                                                                | <b>Cadence</b>                        |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Discovery</b>                         | Session where Contractor shall elicit requirements from Program office stakeholders                               | Cadence to be established by COR      |
| <b>Daily Standup</b>                     | Agile standup                                                                                                     | Daily                                 |
| <b>Release Planning</b>                  | Planning for work to complete in next Release                                                                     | Start of every Release                |
| <b>Sprint Planning</b>                   | Planning for work to complete in next Sprint based on COR goals                                                   | Start of every Sprint                 |
| <b>Product Backlog Refinement</b>        | Adjusting priorities, estimates and details for Product Backlog Items based on direction from COR                 | Once per Sprint                       |
| <b>Sprint Review &amp; Demonstration</b> | Demonstrate work completed in the Sprint                                                                          | Once per Sprint                       |
| <b>Retrospective</b>                     | Review & discussion of what went well and tracking of lessons learned and improvements to apply to future Sprints | End of every Sprint                   |
| <b>Project Roadmap</b>                   | Roadmap that shows project timeline/milestone dates by Release and Epic/User Story based on current estimates     | Updated every Sprint                  |
| <b>Product Backlog</b>                   | Singular source of NHTSA prioritized inventory of all requirements collected by the Contractor                    | Updated every Sprint                  |
| <b>Sprint Backlog</b>                    | Inventory of all work to be completed in the Sprint                                                               | Created every Sprint                  |
| <b>Agile Board</b>                       | Helps plan, visualize and managed the Product and Sprint Backlogs                                                 | Created every Sprint<br>Updated daily |
| <b>Sprint Burndown</b>                   | Daily progress toward work committed in the Sprint                                                                | Updated daily                         |
| <b>Sprint Velocity</b>                   | The team's Velocity (points completed vs. planned) in every Sprint                                                | Updated every Sprint                  |
| <b>Release Burndown</b>                  | Daily progress toward work committed in the Release                                                               | Updated daily                         |

|                                                            |                                                                                                                                                                                            |                                                                                                                    |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Epic Burndown</b>                                       | Daily progress toward work in the Epic                                                                                                                                                     | Updated daily                                                                                                      |
| <b>Initiatives, Epics, User Stories, Tasks and Bugs</b>    | Hierarchy to organize and track requirements/issues                                                                                                                                        | Updated/Reviewed during every Product Backlog Refinement session                                                   |
| <b>Acceptance Criteria</b>                                 | Criteria that define when an issue can be considered complete                                                                                                                              | Integral part of every User Story/Task/Bug                                                                         |
| <b>Level of Effort Estimation in Units of Story Points</b> | Level of effort to complete the work for a User Story/Task/Bug from start to finish (analysis, design, development, test, deployment, documentation, etc.)                                 | Integral part of every User Story/Task/Bug                                                                         |
| <b>Definition of Ready</b>                                 | Detailed criteria that define when a task is truly ready for development (e.g., there are no requirements clarification questions, the task is not blocked waiting for NHTSA action, etc.) | Included in the ASDLCP<br>Updated as necessary or directed by COR<br>Pre-requisite before a task can be started on |
| <b>Definition of Done</b>                                  | Detailed criteria that define when a task is truly done. Applies to all tasks.                                                                                                             | Included in the ASDLCP<br>Updated as necessary or directed by COR                                                  |

*Table 1 - Minimum Set of Agile Events/Deliverables the Contractor shall Conduct/Provide on Cadence*

On a daily basis, the Contractor shall ensure all issues in NHTSA's Agile management tool are accurate and complete. At any given time, for any given Initiative/Epic/User Story, Sprint or the overall project, the Contract must be able to answer the following questions quickly based on the information in the NHTSA Agile management tool: 1) How much work has been completed? 2) How much work remains? 3) Is the item on track for completed as scheduled?

It is extremely important to note that all reports and overall project status are informed by the status of the issues in the NHTSA Agile management tool. The Contractor shall ensure that all built-in and plug-in provided reports within NHTSA's Agile management tool (e.g., Velocity, Release Burndown, Sprint Burndown, Cumulative Flow, Control Chart, etc.) are accurate and up-to-date on a daily basis such that the COR and NHTSA stakeholders can depend on the information they convey. For any given report, if the underlying issues do not have the correct information (summary, status, estimate, etc.), then a garbage-in-garbage-out situation could arise and the accuracy of the report accordingly adversely impacted.

The Contractor shall also ensure that the Agile board is accurate and represents all activities in priority order along with their correct status. The Agile board will be referenced by NHTSA to review the current queue of activities as a key element of the overall project status.

A Sprint will be two or three weeks long, as determined by the COR. The Contractor shall ensure that all changes implemented in a given Sprint-- a product increment-- are ready to be deployed to Production at the end of the Sprint if approved by the COR. The Contractor shall ensure that every product increment provides value to NHTSA within the scope of this SOW. A release will be comprised of two or three product increments, as determined by the COR. Changes to the Production environment (e.g., release deployment) can only be made with the approval of the COR, who will ensure all DOT/NHTSA requirements are met before approving the change. The Contractor is responsible for planning a release in advance, preparing a checklist and all required artifacts to ensure a timely and smooth approval of a Production deployment by the COR.

The Contractor shall ensure that all tickets (Initiative, Epics, User Stories, Tasks and Bugs) in NHTSA's Agile management tool have a sound and consistent structure with succinct but descriptive summaries and comprehensive, well-written details so they can be understood by the widest audience possible and serve as good historical references in the future. The Contractor shall also ensure that every ticket has a level of effort estimation in story points that is driven by team discussion, sound technical approach and consensus.

The Contractor shall also ensure that an Acceptance Criteria is present for every relevant ticket, and which is accurate and comprehensive. For bugs, the contractor shall ensure the ticket clearly provides context for the bug, states the business and technical impacts of the bug, steps to reproduce the problem, why it is a problem to begin with, and any available workarounds. Most importantly, the Contractor shall ensure that no ticket is marked as completed until the Definition of Done is met.

The Contractor shall be responsible for meeting the goals of every Sprint and Release, working with the COR if issues arise to ensure adjustments are made to correct course without impacting the overall cost, scope or schedule of this contract.

#### **4.3.2 Requirements Definition & Management**

The Contractor shall be responsible for creating and maintaining a Requirements Definition Document (RDD) for review and approval by the COR in accordance with the Agile Schedule. The COR will accept or reject the RDD in accordance with the Agile Schedule. The Contractor shall maintain and/or revise the RDD as directed by the COR.

The RDD will serve as the overarching and complete repository of high-level requirements which will be the main source of the Product Backlog, which the Contractor shall decompose into logically organized Use Cases, Initiatives, Epics, User Stories, Tasks and User Story Maps as appropriate. The RDD will anchor the requirements documentation mechanism by providing a starting point for capturing all NHTSA needs within the scope of this SOW as product capability requirements. The Contractor shall ensure that the RDD includes functional and non-functional requirements, data requirements, reporting requirements, external interfaces, dependencies and constraints, as well as standards and technical compliance requirements such as cybersecurity, usability (e.g., Section 508) and performance metrics.

Additionally, as part of Agile planning and decomposing the work to be completed, the COR may request the Contractor to build an Agile Work Breakdown Structure (WBS) as an overarching organization tool for feature completion, releases, Sprints within releases, activities within a Sprint, etc.

While the COR will prioritize requirements and make trade-off decisions, the Contractor shall be responsible for building and maintaining the corpus of requirements in the RDD, and the

decomposed requirements that flow from it, by conducting customer needs analysis and elicitations through one more stakeholder session to be coordinated through the COR.

As requirements change and Product Backlog refinement sessions dictate, the Contract shall update the RDD to reflect the current requirements. However, due to its high-level nature, it is expected the RDD will change infrequently relative to the decomposed, fine-grained user stories, which the Contractor shall create and maintain in NHTSA's Agile project management tool.

#### **4.3.3 Requirements Traceability Matrix (RTM)**

The Contractor shall create and maintain a Requirements Traceability Matrix (RTM) as part of the requirements elicitations and analysis process and submit it to the COR for approval. The COR will accept or reject the RTM in accordance with the Agile Schedule. The Contractor shall maintain and/or revise the RTM as directed by the COR.

The Contractor shall ensure that the RTM clearly links requirements to where and how they have been implemented in the system. The Contractor shall ensure that the RTM provides bidirectional traceability, meaning the RTM documents each requirement from its definition or reference in the RDD through analysis, design, testing, acceptance, and deployment.

#### **4.3.4 Solution Architecture**

The Contractor shall develop/update and maintain the TRIPRS Solution Architecture Document (SAD) to specify the software layers (front end, middle tier, and backend) and the interconnection among major components, including services/APIs, databases information /document stores and third-party software/services/components in support of the requirements provided in this SOW. The COR will accept or reject the SAD in accordance with the Agile Schedule. The Contractor shall maintain and/or revise the SAD as directed by the COR.

The Contractor shall ensure that the SAD includes comprehensive logical and physical architecture diagrams, their descriptions, system data flow diagrams, security components and performance and scalability objectives and designs. The Contractor shall make architectural choices and recommendations that inherently provide cybersecurity, performance, reliability, scalability, extensibility, high efficiency, reduced costs and innovation for NHTSA. The Contractor shall leverage NHTSA approved cloud technologies, managed services and serverless resources, including new and emerging technologies, that allow NHTSA to innovate on behalf of its end users and focus more on solving business problems and less on infrastructure management.

All technical solutions are subject to review by the NHTSA Architectural Review Board (NARB) and will be coordinated by the COR. The Contractor shall present the TRIPRS technical designs with the NARB periodically and as directed by the COR, and address any feedback and findings in a timely manner as directed the COR. The Contractor shall also undergo an Oracle cloud Well-Architected Framework review, coordinated by the COR, and address any feedback and findings in a timely manner as directed by the COR.

#### **4.3.5 Design Specifications**

The Contractor shall develop and submit, to the COR, a Technical Design Specification (TDS) in accordance with the Agile Schedule for all features. The COR will accept or reject the TDS in accordance with the Agile Schedule. The Contractor shall maintain and/or revise the TDS as directed by the COR.

As appropriate, the Contractor shall ensure that the TDS includes the design of the backend, including algorithms, sequencing of calls between all services, synchronous and asynchronous processes, error handling, etc., and how relevant business workflows and rules will be

implemented. The TDS need not address all low-level details, but the Contractor shall ensure the TDS contains sufficient details on the implementation blueprint for a given capability, including key design decisions and the rationale for them. The Contractor shall ensure that the TDS is compliant with and supports the SAD.

As part of the TDS, the Contractor shall include application process flows, business process models, including concept of operations, flow charts, business workflows, activity diagrams, use case diagrams, etc., as appropriate. The TDS shall account for all impacted systems and processes, clearly convey the technical approaches taken and serve as a communications tool to clearly convey the intended implementation to the Contractor's development team as well as NHTSA technical stakeholders.

As appropriate, the Contractor shall also develop and include in the TDS a User Interface Design (UID) that includes wireframes and/or screen mockups that depict how the user interface will be laid out and how UI elements and controls will behave.

The Contractor shall also develop an Interface Control Document (ICD) to describe touchpoints and interactions with other systems. The Contractor shall ensure that the ICD spells out the connectivity, security, communication protocol, data flow, timing, synchronicity, sequencing, data exchange formats and expected Service Level Agreements (SLA) between the TRIPRS system and such systems. The Contractor shall include illustrative diagrams to convey the interaction between the systems and the other systems. The Contractor shall collaborate and coordinate through the COR with the other systems' points of contact to reach agreement on the ICD.

#### **4.3.6 DevSecOps and CI/CD**

The Contractor shall utilize DevSecOps practices to build secure, high quality TRIPRS software in an automated fashion. The Contractor shall build upon the TRIPRS system's existing Continuous Integration (CI) and Continuous Deployment (CD) pipeline to develop and deploy its solutions across all environments: Development, Test, Staging, and Production environments. The Contractor shall automate application code unit test, integration tests, static and dynamic security tests, functional, and regression tests. The Contractor shall enable CD for the TRIPRS systems by automating all the software build, testing and deployment activities and avoiding any manual steps (except possibly COR approval steps).

The Contractor shall ensure all code, scripts, configurations, database changes, etc., are checked into the NHTSA code repository and deployed via the CI/CD pipeline. Additionally, the Contractor shall follow the DOT/NHTSA configuration and change management processes to apply changes to the TRIPRS CI/CD pipeline and application infrastructure (e.g., databases, infrastructure, network, object storage, and compute resources).

The Contractor shall incorporate Static Application Security Testing (SAST), Dynamic Application Security Testing (DAST) and code quality scans as part of the build and deployment process and resolve all findings to meet DOT/NHTSA cybersecurity requirements before deploying solutions to the Staging or Production environment.

Once the application is automatically built, automated functional, regression and load testing run and pass, automated security testing (static and dynamic) run and pass, and UAT passes, the Contractor shall ensure that the deployment to Production proceeds in a controlled and automated fashion with no manual intervention, other than the required approval by the COR to deploy to Production.

The Contractor shall incorporate the DevSecOps approach in the ASDLCP. Further, the Contractor shall review the DevSecOps activities and CI/CD pipelines on an ongoing basis and make enhancements to streamline both, increase automation and integration of the various pipeline stages, reduce errors and decrease build and deployment times.

#### **4.3.7 Testing**

The Contractor shall conduct functional, unit, system/integration, regression, smoke, load/performance and/or stability tests as applicable for each system release. The Contractor shall use industry-standard automated testing approaches and any NHTSA approved tools, such as JMeter for load/performance testing, Cucumber for acceptance testing, and Gauge or Selenium for UI testing. The Contractor shall utilize load/performance testing to stress test the system entry points (APIs, database queries, search queries, etc.) as well as from the user's perspective by simulating concurrent real users accessing the system and carrying out typical use cases. The Contractor shall ensure that load/performance testing includes various load scenarios that the system is likely to experience during peak usage by users.

The Contractor shall ensure at least 85% of all tests in each category are automated and run on a scheduled basis or triggered a set of predetermined criteria or events. The Contractor shall ensure that the full history and results of each run of tests are easily accessible to authorized NHTSA personnel.

The Contractor shall conduct an amount and type of testing commensurate with the size, scope, and risk of the specific release as mutually agreed upon by the Contractor and the NHTSA COR. The Contractor shall develop a Test Plan that outlines the Contractor's overall approach for conducting the various tests mentioned in this section throughout the development process, describes the test environment(s), the process and types of tests to conduct, and the process for recording test findings and sharing test result reports with NHTSA.

The Contractor shall develop and provide a UAT Plan (UATP) for every software release and submit it to the COR for approval at least 5 days prior to the UAT event. The COR will accept or reject the UATP within three (3) days of receipt. The Contractor shall revise the UATP as directed by the COR.

The Contractor shall prepare all UAT test cases in the UATP, set up required data to facilitate the UAT in the UAT environment (typically Staging) and, as appropriate, provide pre-UAT training for users that will participate in the UAT. The Contractor shall ensure that test cases and the UAT activities cover all changes (e.g., user interface, data entry, workflow activities, reporting and other changes that may have operational impact on the system) as well as all user roles.

The Contractor shall ensure that the cybersecurity, access control, performance, and usability of the new system in the UAT environment is identical to how the system will behave in Production.

The Contractor shall facilitate and assist NHTSA in performing UAT when a product increment has passed the Contractor's functional testing, regression testing and quality control checks. Although NHTSA is responsible for conducting the UAT, the Contractor shall facilitate the UAT sessions, document all findings in real time during the UAT and work with NHTSA to prioritize remediation actions and plan additional UAT sessions as appropriate. The Contractor shall ensure all defects and feedback items, including request for new enhancements, are tracked in NHTSA's Agile project management tool.

The UAT is considered passed and successful only when the COR has accepted the product increment which underwent UAT. The Contractor is responsible for obtaining a written product acceptance from the COR, which is required for deploying the product increment in Production.

#### **4.3.8 Software Release Deployment**

The Contractor shall release software into the Production environment on a regular cadence according to the Agile Schedule following the process outlined in the ASDLCP. The Contractor shall create a comprehensive, step-by-step Release Deployment Checklist (RDC) listing all activities for every deployment to Staging and Production and submit it to the COR for approval at least five (5) days prior to the software release date. The COR will accept or reject the RDC within two (2) days of receipt. The Contractor shall revise the RDC as directed by the COR.

The RDC must include assumptions, constraints, dependencies, and contingencies. The Contractor shall identify all parties on which the release has a dependency and clearly identify their tasks in the RDC. The RDC must also contain a rollback plan that the Contractor will execute to revert the system to the pre-deployment state if/when a release deployment proves non-viable and NHTSA approves a rollback. The Contractor shall ensure that the RDC has a mandatory section on cybersecurity and privacy to ensure compliance with the requirements outlined in this SOW. The Contractor shall ensure that no software is deployed, or any changes made in the Production environment without approval from the COR.

The Contractor shall prepare and manage Release Notes to document all changes (fixes, enhancements, and new features) included in each system release and support the release process using the NHTSA standard tools. The Contractor shall submit the Release Notes to the COR prior to every release. The Contractor shall ensure every release is well planned and must collaborate with NHTSA to coordinate support from all required resources within and outside NHTSA. The Contractor shall distribute the RDC and conduct such coordination at least two weeks in advance of every release. The COR will review the RDC and Release Notes in accordance with the Agile Schedule. Once approved by the COR, the Contractor shall disseminate the Release Notes to users and/or stakeholders as directed by the COR.

#### **4.3.9 User Training**

In addition to any in-application help contents and the User Guide, the Contractor shall, for all features, create resources, including documents, slides, sample data, videos, and hold training sessions, with dummy data and hands-on exercises as appropriate, for internal and external end users on newly developed or changed functionality and applications. The Contractor shall update such user training materials as part of every software release and ensure it covers the changes in that release.

The Contractor shall develop and provide to the COR the relevant training materials within five (5) days of the training event or due date. The COR will accept or reject the training materials within two (2) days. The Contractor shall revise the training materials as directed by the COR.

## **5 TASKS**

### **5.1 Project Management**

The Contractor shall manage work in accordance with standard practices derived from the Project Management Institute (PMI) Project Management Book of Knowledge (PMBOK), prominently incorporating Agile delivery methodologies. The Contractor shall develop and submit to the COR

a Project Management Plan (PMP) within forty-five (45) days of award. The Contractor's PMP shall describe the practices the Contractor will employ to manage this project and must include:

- Change management plan
- Communications plan
- Configuration management plan
- Cost and schedule management and control plan
- Detailed deliverables schedules
- Detailed milestone schedules
- Management process
- Methods, tools, and techniques
- Monitoring and control mechanisms
- Organizational structure & resource allocation matrix
- Project charter
- Project schedule & implementation plan
- Quality management plan
- Risk management plan (including risk register for tracking and mitigation)
- Roles and responsibilities
- Status collection and reporting process
- Staffing and onboarding plan
- Technical process

The COR will review and return the PMP to the Contractor with comments and edits or approve the PMP within 30 days of receipt.

#### **5.1.1 Contract Kickoff**

The Contractor shall coordinate with the COR and begin preparatory activities for the Contract Kickoff meeting immediately after award.

Within ten (10) days of award, the Contractor shall attend a Contract Kickoff meeting with the COR and Contracting Officer, and any other interested NHTSA personnel at DOT Headquarters in Washington, D.C or virtually via an online meeting. The purpose of this meeting is to discuss the administration of the Task Order, the project's objectives, planned course of action, milestones and deliverables, the work plan, the Contractor's organizational chart and status for the project, transition-in review and status, its understanding of key stakeholders and their roles, lines of communication, project personnel – including onboarding, security requirements, building access and PIV access, and other issues the Contractor or Government identifies. The Contractor shall conduct a short briefing to NHTSA describing their planned approach. After the briefing, the Contractor, COR, and other NHTSA personnel will discuss specific details of the project to resolve any differences in the approach and/or expected goals, and final work plan.

At least five (5) days prior to the Kickoff Meeting, the Contractor shall provide a Kickoff Meeting Agenda & Presentation Materials for review and approval by the COR prior to finalizing. The COR will review and return the Kickoff Meeting Agenda & Presentation Materials to the Contractor with comments and edits or approve them within two (2) days of receipt.

The Contractor shall draft and provide meeting minutes report within two (2) days after the Kickoff Meeting documenting the Kickoff Meeting discussion and capturing any action items.

### **5.1.2 Weekly Status Report**

At the start of every week, the Contractor shall submit a Weekly Status Report to the COR containing any relevant updates from the previous week and planned items for the upcoming week. The Contractor shall ensure the Weekly Status Report includes, at a minimum, the following information: Contractor's activities regarding DME and O&M, contract finances, issues, risks/mitigations, staffing and cybersecurity.

### **5.1.3 Monthly Progress Reports**

On or before the second Monday of each month following the month being reported, the Contractor shall submit a letter-type Monthly Report that includes, at a minimum, the following information:

1. Accomplishments made during the reporting period.
2. Planned actions and associated costs for the following month.
3. Problems or delays that the Contractor has experienced and specific actions that the Contractor proposes to undertake to alleviate the problems or delays.
4. Preliminary or interim results, conclusions, trends, or other information that NHTSA would need to know.
5. Any actions the Contractor requires of NHTSA.
6. Costs incurred during the reporting period compared with budgeted costs.
7. Total costs incurred to date compared with budgeted costs.
8. Cost breakdown at the project or task level and overall contract for the current month and all prior months, and cost projections at the same levels through the end of the Period of Performance.
9. The percent of work completed for the contract overall and individually by task;
10. Schedule information available at the task level.
11. If the Contractor's response to the above statement is Over Budget, then the Contractor shall explain those conditions which resulted in the Over Budget status and explain what steps it plans to take in order to mitigate, and eliminate, the Over Budget status.
12. Risk register that the contractor maintains with all current known risks and proposed mitigations

Within five (5) days of receipt, the COR will accept or reject the Monthly Progress Report. Rejected reports will be returned to the Contractor for revision.

### **5.1.4 Central Project Artifacts Repository**

The Contractor shall ensure all project artifacts mentioned in this SOW, including the Decision Log, Risk Register, Agile artifacts, O&M artifacts, deliverables and all related documents and work products are located in a centralized location using NHTSA's SharePoint and/or Confluence, are accessible only by those the COR authorizes as read-only or read/write, as appropriate, and are version controlled. The Contractor shall ensure all projects artifacts are highly organized with a reasonable folder structure, consistently and aptly named and easily accessible and navigable by authorized NHTSA personnel. The Contractor shall consider all projects artifacts as living and the Contractor shall ensure all are kept accurate and current to within one month (the Contractor shall ensure ASDLC related artifacts are kept current at least once per software release). The Contractor shall ensure all artifacts undergo a quality control review for accuracy, completeness and currency as part of the monthly update cadence.

### **5.1.5 Meetings**

In the course of performing the activities required under the scope of this BPA Call, the Contractor shall coordinate and hold meetings as needed to facilitate requirements analysis an elicitation, review certain artifacts and procedures, troubleshoot problems, address risks and mitigations, discuss Contractor recommendations regarding technologies, tools and procedures, or a variety of other issues impacting the TRIPRS system/s.. For all such meetings, the Contractor shall capture Meeting Notes and track action items. The Contractor shall record Meeting Notes and action items in real-time during every meeting and share them immediately upon conclusion of each meeting as a draft. The Contractor shall place all Meeting Notes in the Central Project Artifacts Repository. The Contractor shall ensure that all Meeting Notes are easy to search and find by ensuring they are well-organized, categorized by meeting name and include the meeting date and time, attendees and comprehensive but succinct notes, decisions to be added to the Decision Log, action items with owners and next steps. The Contractor shall track all open action items from all Meeting Notes and follow up with the action item owners to resolution.

### **5.1.6 Decision Log**

The Contractor shall document and maintain all project related decisions in a Decision Log. The Contractor shall ensure every meeting and communication in which a decision impacting project scope, cost or schedule is made is included in the Decision Log. Each decision must be uniquely identified and comprehensively documented to include at least a synopsis, context and details, points of contact involved in the discussions or decision making, timelines, alternatives considered, recommendations, references to relevant resources, such as emails, documents and other internal or external artifacts, and the rationale for the decision. The Contractor shall ensure the Decision Log is located prominently within the Contractor's central project artifacts repository for easy access by authorized NHTSA personnel. The Contractor shall ensure the Decision Log is kept up to date on a weekly basis. The Contractor shall ensure the Decision Log is included in Monthly Progress Reports provided by the Contractor.

### **5.1.7 Risk Register**

The Contractor shall document and maintain all project related risks in a Risk Register. The Contractor shall ensure the Risk Register is located prominently within the Contractor's project central artifact repository for easy access by authorized NHTSA personnel. The Contractor shall ensure the Risk Register is kept up to date on a weekly basis. The Contractor shall ensure the highest Risk Register items, as determined by the COR, are included in Monthly Progress Reports provided by the Contractor.

### **5.1.8 Standard Operating Procedures**

For activities that are repeatedly applied for routine work in the execution of the requirements within the scope of this SOW, the Contractor shall develop and maintain Standard Operating Procedures (SOPs). For example, personnel onboarding and offboarding procedures, Helpdesk procedures, etc. The Contractor shall ensure its SOPs are located prominently within the Contractor's project central artifact repository for easy access by authorized NHTSA personnel. The Contractor shall ensure the SOPs are kept up to date on a monthly basis.

## **5.2 Development, Modernization and Enhancements (DME)**

The Contractor shall develop a comprehensive set of TRIPRS capabilities to facilitate NHTSA's end-to-end workflows. The Contractor shall analyze the requirements, develop and document detailed requirements and RTM using the NHTSA Agile tools; and design, develop and deploy

each of the capabilities as directed by the COR. In the following sections, the requirements for some capabilities are more detailed than others, owing to previous analysis of the capability within NHTSA. In all cases, the requirements provided here must be considered initial and the Contractor shall be prepared to further analyze and each capability more carefully and in more detail they are implementing iteratively following Agile methods as required for this project. Additionally, because the Government is utilizing Agile methods, these may not represent all of the requirements. The Contractor shall be prepared to conduct discovery and elicitation and continuously adapt as part of the Agile process to better understand NHTSA stakeholder needs and incrementally implement the right product for NHTSA.

#### **5.2.1 User Management**

Access to the TRIPRS system is by invitation only. The current TRIPRS internal application allows NHTSA personnel to invite external users. The current TRIPRS external application has basic user management features available to designated administrators of external entities to manage TRIPRS access for their Registered Importer users.

The Contractor shall develop user management features which are available to users with user administration permissions. The Contractor shall ensure that the user management capability provides Create, Read, Update and Delete (CRUD) operations for administrators. Examples include inviting users to register, assigning users permissions, changing user profiles and activating/deactivating users. The Contractor shall ensure that the user management capability also provides the ability to list users and easily search for users by various attributes, whole or partial value of key user attributes, such as the user's status, organization, any part of the user's name or email address, the ability to page through the result set and the ability to set the number of results per page. The Contractor shall ensure that NHTSA administrators can fully manage both internal and external users.

The Contractor shall ensure that approved active users are able to log into the associated TRIPRS application to view their account details, manage, as appropriate, aspects of their profile (organization, name, email address, phone, etc.), and view their activities, such as their last login date and time, any application preferences, etc.

The Contractor shall ensure that all changes to a user's profile trigger notifications to the appropriate parties describing the nature of the change and are included in the audit trail and available to authorized TRIPRS internal and external users to view.

Because external entities may use common third-party entities to handle some or all of their requests for confidential treatment, the Contractor shall devise a viable solution for treating the relationship of such users, representing multiple entities, as separate while recognizing the user account itself, identified with a unique email address, as singular.

The Contractor shall, with guidance from NHTSA, build upon or redevelop the current user permissions model (roles/permissions, association with external entities and relationships to each other) as appropriate to ensure they are comprehensive, easy to understand, are fine grained and provide clear descriptions and associations with the features for which they are intended. The Contractor shall collaborate with other NHTSA external entities that need to provide external user management via the NHTSA Enterprise Portal to ensure a common design and reuse of architecture, design components and code, or make this a wholly independent application that can be customized to handle multiple applications with their own roles and permissions.

#### **5.2.1.1 Internal User Management**

The Contractor shall ensure that all internal users have self-service feature to register to access the TRIPRS internal applications. The Contractor shall ensure a simple user registration workflow is implemented with automated notifications as appropriate to facilitate self-registration and registration approval by NHTSA TRIPRS program office team. The Contractor shall leverage the DOT user directory to automatically pull the user's information and require minimal input from the user (e.g., why they need access) for the self-registration.

#### **5.2.1.2 External User Management**

Each Registered Importers holds an account with its own set of users that are responsible for their organization.

#### **5.2.2 Administration Console**

Within the TRIPRS internal applications, the Contractor shall provide an Administration Console. The Administration Console will allow TRIPRS Administrators to configure certain application values and control certain application behavior. For example, the Administration Console will allow TRIPRS Administrators to manage external entities that participate in the TRIPRS system. The TRIPRS system must allow NHTSA users to be self-sufficient as possible whenever possible. Depending on frequency and cost, not every single backend change may be suitable for implementing as a feature in the Administration Console. The COR will direct the Contractor on specific functions and scenarios accordingly.

#### **5.2.3 Audit Trail**

The Contractor shall ensure every action any user takes in the TRIPRS system, including all edit actions and certain non-edit actions, is recorded as an audit record. The Contractor shall ensure the audit record is fine-grained and captures field level changes. Thus, an authorized user viewing an audit trail record must be able to tell the user that modified a field, the old value of the field, the new value of the field and the date and time stamp of the change.

#### **5.2.4 Existing Product Backlog**

The TRIPRS systems has an evolving backlog containing various bugs and system enhancements to already implemented capabilities. These backlog items are within the scope of this SOW. The Contractor shall lead efforts to help the COR groom and refine these backlog items on a biweekly basis. As part of the refinement activities, the Contractor shall capture new enhancement requests from the COR and track them using NHTSA's Agile project management tool and seek COR approval to, as appropriate, mark existing items as obsolete or redundant, and consolidate or split them. As described in the Agile Process section, each item in the Product Backlog must be sufficiently detailed and complete and the Contractor shall generally maintain a clean, up-to-date and accurate Product Backlog.

### **5.3 Operations and Maintenance (O&M)**

The Contractor shall develop and submit to the COR an Operations and Maintenance Plan (O&MP) detailing how it will conduct O&M for the TRIPRS system within thirty (30) days of award. The O&MP shall include the Contractor's plan for corrective, preventive and predictive maintenance, as well as observing and monitoring the TRIPRS system. As new features are developed and deployed, the Contractor shall incorporate any special handling for sustainment of that feature in the O&MP. The goal of the O&MP is to maximize the effectiveness and success of the O&M team. The Contract shall ensure the O&MP includes how-to's, SOPs, troubleshooting

tips, technical artifacts and generally all useful information to allow the O&M team and the Helpdesk to find all TRIPRS information quickly. The Contractor shall provide information in the O&MP on data and document backup and recovery/restore procedures as well as applying the TRIPRS system data retention policy (e.g., deleting documents that are older than the NHTSA approved retention period).

The Contractor shall be responsible for the operation and maintenance of all AWS, Oracle services and resources impacting the TRIPRS system, including databases, application software, tools, and software components. The Contract shall apply all changes, updates and upgrades mandated by the COR and DOT/NHTSA in support of security data calls, POA&M resolutions, ATO renewal or other directives.

The Contractor shall ensure that all TRIPRS system tooling and technology stack components are kept current to within one (1) major version of the latest non-obsolete, DOT/NHTSA approved version. The Contractor shall ensure that deprecated features in AWS and/or any component of the technology stack are resolved well in advance of obsolescence. NHTSA or the COR may at any time change this guidance (typically due to cybersecurity policies) and the Contractor shall take necessary actions to ensure the TRIPRS system complies with updated guidance and/or direction from the COR direction as appropriate.

The Contractor shall utilize an Agile framework, such as Kanban, as the core method by which to triage, track, prioritize and address maintenance items.

The Contractor shall ensure that an O&M Agile board accurately representing all O&M activities in priority order is maintained in NHTSA's Agile project management tool. The Contractor shall ensure that every O&M issue, including TRIPRS Helpdesk activities, has a corresponding ticket in the Agile project management tool. The O&M board will be referenced by NHTSA stakeholders to review the current queue of O&M activities. At any time, the COR may direct the Contractor to shift priorities and/or change/add/remove items from the O&M Agile board, and the Contractor shall immediately comply.

### **5.3.1 Records Management**

The Contractor shall comply with NHTSA's record retention and disposition schedules for data and documents for TRIPRS. The Contractor shall implement automated means in the system to comply with the records management policy, for example, deleting documents that have reached or exceeded their retention period.

### **5.3.2 Cybersecurity and Privacy**

The Contractor shall ensure that the TRIPRS system architecture adheres to Federal, DOT and NHTSA cybersecurity mandates, guidelines and standards and complies with DOT policies for securing systems and applications, including access and all interactions with the system.

The Contractor shall ensure that cybersecurity is an integral part of all TRIPRS system maintenance activities. The Contractor shall ensure access to the TRIPRS system and data are maximally secured and protected. The Contractor shall ensure that both the internal and external TRIPRS applications allow only Multifactor Authentication (MFA) for users. Additionally, the Contractor shall ensure that data at rest as well as in transit within the TRIPRS systems and other systems is encrypted. The Contract shall ensure that cybersecurity and privacy prerequisites are prominently included as first-class requirements in all phases and aspects of the Contractor's O&M

support activities, technical solutions, software development and general IT maintenance activities.

The Contractor shall ensure that any transfer and exchange of data within the TRIPRS system and applications outside the DOT/NHTSA enterprise is clearly identified and reviewed and approved in writing by the COR, Program offices and NHTSA's Office of IT Compliance and Chief Information Security Officer (CISO).

#### **5.3.2.1 Cybersecurity Data Calls**

The Contractor shall ensure that the TRIPRS system retains a strong security posture by mitigating and remediating all compliance and vulnerability findings within the thresholds defined by OST and NHTSA. The Contractor shall review, properly interpret, and mitigate/remediate compliance and vulnerability findings from various system configuration and security scanners (SAST, DAST, etc.) as appropriate.

#### **5.3.2.2 Security Authorization Package, Authorization to Operate and Continuous Monitoring**

The Security Authorization Package (SAP) for NHTSA projects is based on the National Institute of Standards and Technology (NIST) Risk Management Framework (RMF) as well as related NHTSA and DOT standards, guidelines, policies and directives. On an annual basis, the SAP documents the results of the security controls assessment and provides the Authorizing Official (AO) with essential information needed to make an informed risk-based decision on whether to authorize the operation of an information system. Upon approval by the AO, the SAP facilitates the Authorization to Operate (ATO), which allows the project to operate.

Currently, the TRIPRS system is within the security boundary of another NHTSA project. As such, its SAP artifacts are merged with the other system. It is likely that the TRIPRS systems will be split out into its own security boundary for purposes of the ATO. The Contractor shall support all SAP activities, including continuous monitoring of security controls and annual ATO renewal tasks, by providing information asked of them, creating or updating required documentation, and implementing all cybersecurity system updates and process changes as directed by the NHTSA CISO. The Contractor shall ensure that its staff have the expertise to independently address and complete all required SAP artifacts, questions and activities promptly, accurately, comprehensively and with high quality. The Contractor shall coordinate with the COR to ensure it receives input from all required parties in making its updates to the SAP. Some of the SAP requirements call for associated activities to exercise or periodically validate their contents. For example, the Information System Contingency Plan (ISCP) requires a validation exercise; thus, the Contractor shall conduct a yearly exercise to validate the ISCP. The Contractor shall support all such SAP related activities. The Contractor shall fully support all cybersecurity and ATO inspection and assessment activities conducted on the TRIPRS system by DOT or NHTSA sanctioned entities.

The Contractor shall ensure the SAP is located prominently within the Contractor's central project artifacts repository for easy access by authorized NHTSA personnel. The Contractor shall ensure the SAP is kept up to date on a monthly basis. The Contractor shall submit an accurate and up-to-date SAP to the COR for review on request. The Contractor shall submit the SAP artifacts to the COR ninety (90) days before the ATO recertification or renewal date. The COR will review and return the SAP artifacts to the Contractor with comments and edits or approve the SAP artifacts within thirty (30) days of receipt.

Regardless of the selected controls for the SAP for a given year, the Contractor shall keep all security control artifacts for which responsibility is partly or wholly on the TRIPRS project team complete and accurate throughout the period of performance.

All security controls are derived from the NIST special publications and adopting/supporting DOT and NHTSA policies and guides. Examples of key security control artifacts the Contractor is responsible for include:

- Configuration validation checklist
- Disaster recovery plan (DRP)
- Incident response plan (IRP)
- Information security continuous monitoring (CM) plan
- Information system configuration management (ISCM) plan
- Information system contingency plan (ISCP)
- Plan of action and milestones (POA&M)
- Ports, protocols, and services (PPS) list
- Privacy impact assessment (PIA)
- Privacy threshold assessment (PTA)
- Software/hardware/cloud resources/cloud managed services inventory list
- System architecture/boundary/topology/data flow
- System data backup and recovery (SDBR) plan
- System security plan (SSP)
- Security-focused configuration management (SecCM) plan

## **5.4 Transition**

### **5.4.1 Transition-In**

The Contractor shall develop and submit to the COR a draft Transition-In Plan within five (5) days of award, and a final Transition-In Plan with fifteen (15) days of award. The Contractor shall ensure that the Transition-In Plan includes information on how the Contractor plans to begin operations to execute the requirements within the scope of this SOW without any impact to NHTSA operations.

The Contractor shall coordinate O&M transition activities with the existing O&M contractor, through the COR, in accordance with the Transition-In Plan. There will not be a transition-in period for any non-O&M tasks because there will be no ongoing incumbent activities for non-O&M efforts.

### **5.4.2 Transition-Out**

For the final two months of the overall BPA Call's period of performance, the Contractor for this project shall transition to a potentially new Contractor. The Contractor shall utilize the transition period to transfer knowledge from the incumbent to the incoming Contractor. The Contractor shall train the incoming personnel. The Contractor shall submit to the COR a draft Transition-Out Plan sixty (60) days prior to the end of the BPA Call, and a final Transition-Out Plan thirty (30) days prior to the end of the BPA Call. The Transition-Out Plan shall include, but is not limited to:

- Contractor's transition team organization, transition manager and key points of contact
- Transition roles and responsibilities

- Transition milestones
- Transition communication plan
- Inventory of the Contractor's SOPs
- Inventory of accounts, servers certificates, hardware, software, and URLs
- Inventory of licensed products used by the Contractor's team in support of NHTSA
- Documentation – An inventory of all project artifacts from the Contractor's central project artifacts repository with descriptions of each
- Knowledge Transfer – Overview sessions and question and answer sessions via in person meetings, virtual meetings, phone calls, emails, chat, etc. for the incoming Contractor to cover:
  - Review of system functionality, technical system architecture, interfaces and data flow
  - All the Contractor's central project artifacts repository
  - Review of all SOPs
  - Answers questions from the incoming Contractor
- Plan for offboarding the Contractor's staff, including an inventory and accounting of Government Furnished Equipment (GFE) and tokens, system accounts and user accounts
- Plan for ensuring the Contractor executes the transition-out plan without impacting to NHTSA/ODI operations

Prior to submitting the Transition-Out Plan to the COR, the Contractor shall complete a comprehensive review and update, as necessary, of all project artifacts. The Contractor shall ensure all project artifacts mentioned in this SOW, including the Decision Log, Risk Register, Agile artifacts, O&M artifacts, deliverables and all related documents and work products located in the central project artifacts repository are accurate, complete and up to date.

## 5.5 Compliance with the Rehabilitation Act

The Contractor shall be required to ensure that electronic documents it prepares will meet the requirements of Section 508 of the Rehabilitation Act. The Rehabilitation Act requires that all electronic products prepared for the Federal Government be accessible to persons with disabilities, including those with vision, hearing, cognitive, and mobility impairments. When an electronic document developed under this effort have been identified by NHTSA as necessary and required to meet the requirements of Section 508, it is understood that all reasonable costs incurred by the Contractor that are associated with meeting the requirements for Section 508 compliance with be fully reimbursed by the Government.

The Contractor should review Section 508 of the Rehabilitation Act <https://www.access-board.gov/guidelines-and-standards/communications-and-it/about-the-section-508-standards/section-508-standards> and the Federal IT Accessibility Initiative (home page) <http://section508.gov> for further information on these requirements.

## 5.6 ACRONYMS

|     |                       |
|-----|-----------------------|
| BI  | Business Intelligence |
| CCB | Change Control Board  |

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| <b>CO</b>         | Contracting Officer                                      |
| <b>COR</b>        | Contract Officer Representative                          |
| <b>DOT</b>        | U.S. Department of Transportation                        |
| <b>MMUCC</b>      | Model Minimum Uniform Crash Criteria                     |
| <b>NDR&amp;TR</b> | National Driver Register and Traffic Records (Division)  |
| <b>NHTSA</b>      | National Highway Traffic Safety Administration           |
| <b>OCIO</b>       | Office of the Chief Information Officer                  |
| <b>O&amp;M</b>    | Operations and Maintenance                               |
| <b>PM</b>         | Project Manager                                          |
| <b>SME</b>        | Subject Matter Expert                                    |
| <b>SPA</b>        | Safety Program Assessment                                |
| <b>STRAP</b>      | State Traffic Records Assessment Program (System Module) |
| <b>T&amp;M</b>    | Time and Materials                                       |
| <b>TR</b>         | Traffic Records                                          |
| <b>TRIPRS</b>     | Traffic Records Improvement Program Reporting System     |